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Sonargaon University (SU)
সোনারগাঁও ইউনিভার্সিটি (এসইউ)

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LECTURE SHEET

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Chapter: Cost of Capital

Questions at a glance:

1. Define cost of Capital.
2. Describe the importance of Cost of Capital.
3. What are the components of cost of capital?
4. Why is the cost of debt capital calculated on an after-tax basis?
5. Advantages of using debt as capital structure.
6. What about retained earnings? Why retained earnings is usually a cheaper source of equity financing than making a new issue of ordinary shares?
7. Define Weighted Average Cost of Capital (WACC).
8. What factors that affect the Weighted Average Cost of Capital (WACC)?

1. *Define cost of Capital.*

The cost of capital can be defined as, the minimum rate of return a firm is required to earn on its investments in order to satisfy investors and to maintain its market value. In other words it is the investor's required rate of return.

2. *Describe the importance of Cost of Capital.*

Computation of cost of capital is a very important part of the financial management to decide the capital structure of the business concern. The importance of cost of capital is given below:

1. Importance to Capital Budgeting Decision: Capital budget decision largely depends on the cost of capital of each source. According to net present value method, present value of cash inflow must be more than the present value of cash outflow. Hence, cost of capital is used to capital budgeting decision.

2. Impm1ance to Structure Decision: Capital structure is the mix or proportion of the different kinds of long term securities. A firm uses particular type of sources if the cost of capital is suitable. Hence, cost of capital helps to take decision regarding structure.

3. Importance to Evolution of Financial Performance: Cost of capital is one of the important determining which affects the capital budgeting, capital structure and value of the firm. Hence, it helps to evaluate the financial performance of the firm.

4. Importance to other Financial Decisions: Apart from the above points, cost of capital is also used in some other areas such as, market value of share, earning capacity of securities etc. hence; it plays a major part in the financial management.

3. *What are the components of cost of capital?*

The components of cost of capital are given below:

1. Cost of Debt: Debt includes all interest bearing securities. Cost of debt is usually shown as the interest rate charged by the lender. It does not take account of taxes; it is a pre-tax number. Debt is cheaper than preferred stock and common stock because of the tax factor. Formula for calculating cost of debt –

$$\text{Before-tax cost of debt, } k_d = \frac{I + \frac{(\$1{,}000 - N_d)}{n}}{\frac{N_d + \$1{,}000}{2}}$$

$$\text{After-tax cost of debt, } k_{dt} = k_d(1-T)$$

2. Cost of Preferred Stock: The cost of preferred stock may be estimated by dividing the expected (or past) preferred dividend per share by the expected price of the preferred stock per share (or the total preferred dividend by the total value of the preferred stock). Formula for calculating cost of preferred stock –

$$\text{Cost of preferred stock, } k_p = \frac{D_p}{P_0 - F}$$

3. Cost of Equity: A firm's cost of equity represents the compensation that the market demands in exchange for owning the asset and bearing the risk of ownership. The traditional

formulas for cost of equity are-

$$\text{Cost of equity, } k_e = \frac{D_1}{P_0 - F} + g$$

4. *<u>Why is the cost of debt capital calculated on an after-tax basis?</u>*

The after-tax cost of debt is calculated as $k_d(1 - T)$, where the before-tax cost of debt is k_d and T is the marginal rate of tax. The before-tax cost of debt, k_d is adjusted to reflect the tax relief available on debt interest. As debt interest is a tax deductible expense, the cost of debt capital to the firm will be reduced by the amount of tax relief currently available.

For example, if the tax rate is 30 per cent and a firm is paying investors 10 per cent annual interest on £100 par value bonds, then the annual cash flow to investors will be £10 on each bond. However, the firm will save £3.00 (£10×0.30) each year in tax payments.

Over a number of years this tax saving, also known as the tax shield, can be substantial. This assumes of course that the firm has sufficient taxable profits to be able to take full advantage of the tax relief.

5. *<u>Advantages of using debt as capital structure.</u>*

Companies often use debt when constructing their capital structure, which helps lower total financing cost. In addition to the relatively lower cost of debt financing, using debt has other advantages compared to equity financing that are given below:

1. Cost Reduction: Compared to equity, debt requires lower financing cost. Thus, companies often mix debt into their capital structure to bring down the average financing cost. Using debt, companies are contractually liable to make periodic interest payments and return debt principal at maturity. As a result, debt holders bear less risk, compared to equity holders.

2. Profit Retention: While using debt may add pressure to a company's ongoing operations as a result of having to meet interest-payment obligations, it helps retain more profits within the company compared to using equity, which requires the sharing of company profits with equity holders.

3. Leverage: Using debt is also advantageous to existing owners because of the effect of financial leverage. When companies use debt to provide addition capital for their business operations, equity owners get to keep any extra profits generated by the debt capital, after any interest payments.

4. Tax Savings: Using debt helps lower a company's taxes because of allowable interest deductions. Tax rules permit interest payments as expense deductions against revenues to arrive at taxable income. The lower the taxable income, the less taxes a company pays. On the other hand, dividends paid to equity holders are not tax-deductible and must come from after-tax income. Therefore, tax savings help further reduce a company's debt financing cost, which is an advantage that equity financing lacks.

6. *What about retained earnings? Why retained earnings is usually a cheaper source of equity financing than making a new issue of ordinary shares?*

Retained earnings are the proportion of the firm's after-tax distributable profits which have not been paid out to shareholders in the form of dividends. They have been reserved within the firm to finance future investment plans.

Shareholders will expect the firm's managers to earn a satisfactory rate of return on retained earnings in order to at least maintain the market value of the firm's equity; otherwise they should be paid out as dividends so that the shareholders can invest these funds for themselves. Retained earnings therefore incur an opportunity cost.

Retained earnings are part of shareholders' funds or equity; they belong to the shareholders and will appear in the equity part of a firm's balance sheet under capital and reserves. An increase in retained earnings will increase shareholder equity, and in this respect retained earnings are similar to fully paid up ordinary share capital. Therefore the cost of retained earnings is equal to the rate of return required by investors on the firm's ordinary share capital.

We can say that the cost of retained earnings, $k_{re} = k_e$; where, k_e = cost of equity

Retained earnings is usually a cheaper source of equity financing than making a new issue of ordinary shares for following reasons:

i. A new issue would probably have to be made at a discount to the firm's current share price to attract investors (raising the required return on the new issue).
ii. Any new issue would also incur flotation costs (e.g. advertising, brokerage and underwriting fees) of about 3–4 per cent of the total funds raised, reducing the proceeds from the sale.

7. *Define Weighted Average Cost of Capital (WACC).*

The firm's overall cost of capital is traditionally viewed in terms of a weighted average cost of capital (WACC), which is denoted as k_a. It is a simple weighted average of the cost of the individual capital components (equity, retained earnings, preference shares, and after-tax debt).

Weighted average cost of capital (WACC) is calculated as follows:

$$k_a = w_d k_d (1 - T) + w_p k_p + w_e k_e$$

Where,

k_a = weighted average cost of capital

w_d = weight (proportion) of debt in capital structure

k_d = before-tax cost of debt

T = marginal tax rate

w_p = weight (proportion) of preference shares in capital structure

k_p = cost of preference share capital

w_e = weight (proportion) of equity in capital structure

k_e = cost of equity capital

8. *What factors that affect the Weighted Average Cost of Capital (WACC)?*

The weighted cost of capital is affected by some factors that are under a firm's control and some that are not.

A. External Factors: Factors the firm cannot control -

a. The level of Interest rates: If interest rates in the economy rise, the cost of debt increases because firms will have to pay bondholders a higher interest rate to obtain debt capital. Higher interest rates also increase the costs of common and preferred equity.

b. Market Risk Premium: Investors' aversion to risk determines the market risk premium. Individual firms have no control over the MRP, which affects the cost of equity and thus the WACC.

c. Tax Rates: Tax rates, which are influenced by the states, have an important effect on the cost of capital. They are used when we calculate the after-tax cost of debt for use in the WACC. In addition, the lower tax rate on dividends and capital gains than on interest income favors financing with stock rather than bonds.

B. Internal factors: Factors the firm can control:

a. Capital Structure Policy: A firm can change its capital structure, and such a change can affect its cost of capital. For example, the after-tax cost of debt is lower than the cost of equity, so if the firm decides to use more debt and less common equity, then this increase in debt will tend to lower the WACC. However, an increased use of debt will increase the risk of debt and the equity, offsetting to some extent the effect due to a greater weighting of debt.

b. Dividend Policy: The percentage of earnings paid out in dividends may affect a stock's required rate of return. Also, if the payout ratio is so high that the firm must issue new stock to fund its capital budget, then the resulting flotation costs will also affect the WACC.

c. Investment Policy: When we estimate the cost of capital, we use as the starting point the required rates of return on the firm's outstanding stock and bonds, which reflect the risks inherent in the existing assets. Therefore, we are implicitly assuming that new capital will be invested in assets with the same degree of risk as existing assets. This assumption is generally correct, because most firms do invest in assets similar to those they currently use. However, the equal risk assumption is incorrect if a firm dramatically changes its investment policy.

Formula

- **Cost of redeemable debt**

  Before-tax cost of debt, $k_d = \frac{I + \frac{(\$1{,}000 - N_d)}{n}}{\frac{N_d + \$1{,}000}{2}}$

  Alternatively, Using the Hawinini and Vora approximation:

  Before-tax cost of debt, $k_d = \frac{I + [(P - N_d)/n]}{P + 0.6\,(N_d - P)}$

  Where,

  I = annual interest or coupon rate

  P = par value of bond at maturity

  N_d = net proceeds from sale, or market value of debt instrument

  n = number of periods (years) to maturity

  After-tax cost of debt, $k_{dt} = k_d(1-T)$

- **Cost of irredeemable debt**

  Before-tax cost of debt, $k_d = \frac{I}{BV_0}$

  Where,

  k_d = cost of irredeemable bond

  I = annual interest or coupon payment

  BV_0 = current market value of the bond

  After-tax cost of debt, $k_{dt} = k_d(1-T)$

- **Cost of preference share capital**

  $k_p = \frac{D_p}{P_0 - F}$

  Where,

  k_p = cost of preference share capital

  D_p = fixed dividend payment

  P_0 = the current market value of the preference shares

  F = flotation cost

- **Cost of equity capital**
  - Using constant dividend growth model:

    *Cost of retained earnings, $k_e = \frac{D_1}{P_0} + g$

*Cost of new issue, $k_e = \frac{D_1}{P_0 - F} + g$

Where,

k_e = cost of equity

D_1 = expected dividend per share one year from now = $D_o(1 + g)$

P_0 = current market price of the share

g = constant growth rate

F = flotation cost

- Using capital asset pricing model (CAPM)

$k_e = R_f + (R_m - R_f)\,\beta$

Where,

R_f = risk-free rate of return

β = beta coefficient

R_m = expected market return

- Proportion of source of capital = $\frac{\textbf{Available of source of capital}}{\textbf{Total capital}}$

- Break Point = $\frac{\textbf{Available of source of capital}}{\textbf{Proportion of source of capital}}$

- **Weighted average cost of capital (WACC)**

$k_a = w_d k_{dt} + w_p k_p + w_e k_e$

Where,

k_a = weighted average cost of capital

w_d = weight (proportion) of debt in capital structure

k_{dt} = after-tax cost of debt

w_p = weight (proportion) of preference shares in capital structure

k_p = cost of preference share capital

w_e = weight (proportion) of equity in capital structure

k_e = cost of equity capital

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Problems & Solution

1. Weather All Engineering, a mechanical and electrical engineering company, has in issue £100 par value bonds with a coupon rate of 12 per cent paid annually. The bonds, which are currently trading at £93, are redeemable at par in 20 years time. Find the cost of the bond assuming a taxation rate of 30 per cent.

Answer:

Given that,

$P = £100$

$I = £100 \times 12\% = £12$

$N_d = £93$

$n = 20$

Using the Hawinini and Vora approximation:

Before-tax cost of debt, $k_d = \frac{I + [(P - N_d)/n]}{P + 0.6\,(N_d - P)}$

$= \frac{£12 + [(£100 - £93)/20]}{£100 + 0.6\,(£93 - £100)}$

$= \frac{£12.35}{£95.8}$

$= 12.89\%$

After-tax cost of debt $= k_d(1-T)$

$= 12.89\%\ (1 - 0.3)$

$= 9.02\%$

2. Weather All Engineering has in issue 8 per cent irredeemable bonds with a par value of £100. If the market value of a bond is currently £90 and the rate of corporation tax is 30 per cent, find the cost of the bond.

Answer:

Given that,

$I = £100 \times 8\% = £8$

$BV_0 = £90$

Before-tax cost of debt, $k_d = \frac{I}{BV_0}$

$= \frac{£8}{£90}$

$= 8.89\%$

After tax cost of debt $= k_d(1-T)$

$= 8.89\%\ (1 - 0.3) \quad = 6.22\%$

3. Weather All Engineering, a mechanical and electrical engineering company, has in issue 5 million £1.00 par value irredeemable 6 per cent preference shares. Find the cost of the shares if their current market value is £0.96.

Answer:

Given that,

D_p = £1.00×6% = £0.06

P_0 = £0.96

Cost of preference share capital, $k_p = \frac{D_p}{P_0 - F}$

$= \frac{£0.06}{£0.96}$

= 6.25%

4. The directors of Sunny Sky Ventures estimate the dividend for next year at £0.21 per share and expect future dividends to grow at a constant 5 per cent per year. Assuming the current market price of the share is £3.00; find the firm's cost of equity.

Answer:

Given that,

D_1 = £0.21

P_0 = £3.00

g = 5% = 0.05

Cost of equity, $k_e = \frac{D_1}{P_0 - F} + g$

$= \frac{£0.21}{£3.00} + 0.05$

= 0.07 + 0.05

= 12%

5. The dividend per share (DPS) paid by the directors of Carolynco Textile Systems over the period 1994 to 1999 is presented below.

Year	Dividend per Share (DPS)
1999	£0.44
1998	£0.41
1997	£0.38
1996	£0.35
1995	£0.32
1994	£0.30

The directors expect future dividends to continue to grow at the same constant rate per year. Assuming the current market price of the share is £6.30; calculate the firm's cost of equity. Ignore

taxation.

Answer:

We know that,

$D_1 = D_o(1 + g)$

Now, $D_{1999} = D_{1994}\ (1 + g)^5$

or, £0.44 = £0.30 $(1 + g)^5$

or, £0.44/£0.30= $(1 + g)^5$

or, $(1 + g)^5 = 1.467$

or, $1 + g = \sqrt[5]{1.467}$

or, $1 + g = 1.08$

or, g = 1.08 - 1

or, g = 0.08

Here,

$D_1 = D_{1999}(1 + g)$

= £0.44 (1 + 0.08)

= £0.4752

P_0 = £6.30

Cost of equity, $k_e = \frac{D_1}{P_0 - F} + g$

$= \frac{£0.4752}{£6.30} + 0.08$

= 0.0754 + 0.08

= 15.54%

6. The financial manager of Sunny Sky Ventures wishes to use the capital asset pricing model to determine the firm's cost of equity capital. From contacts at the firm's investment bank the financial manager has learned that the risk free rate is currently 6 per cent, the market return is 11 per cent, and the firm's beta is 1.2. Using this information calculate the cost of equity.

Answer:

Given that,

R_f = 6%

R_m = 11%

β =1.2

Cost of equity, $k_e = R_f + (R_m - R_f)\ \beta$

= 6% + (11% - 6%) 1.2

= 12%

7. The financial manager of Sunny Sky Ventures now wishes to determine the company's weighted average cost of capital (WACC) and has collected the following information. The company currently has 5 million shares in issue, no new issues have been made and the current market price is quoted at £3.00. The company also has 50,000 £100 par value bonds issued in the market and which are currently trading at £98, with a current yield of 10 per cent. The cost of equity has been calculated at 12 per cent and corporation tax is 30 per cent. Determine the current weighted average cost of capital (WACC) for the company.

Answer:

The first step is to calculate the respective market values of debt and equity capital and then the respective weightings or proportions of each in the capital structure, thus:

Source	Market value	Weight
Equity (5m×£3.00)	£15.0m	0.75
Debt (50,000×£98)	£4.9m	0.25
	£19.9m	1.00

Weighted average cost of capital (WACC),

$$k_a = w_d k_d (1 - T) + w_e k_e$$

$$= 0.25(10\%)\ (1–0.3) +0.75(12\%)$$

$$=1.75\%+9\%$$

$$=10.75\%$$

8. The Manx Company was recently formed to manufacture a new product. It has the following capital structure in market value terms:

13% Debenture	tk. 6000,000
Preferred Stock	tk. 2000,000
Common Stock (320,000 Shares)	tk. 8000.000
	Total = tk. 16000,000

The company has a marginal tax rate of 40 percent. A study of publicity held companies in this line of business suggests that the required return on equity is about 17 percent. The Manx Company's preferred stock is yielding 12 percent. Compute the firm's weighted average cost of capital.

Answer:

Given that,

Before-tax Cost of Debt, k_d = 13%

Cost of preferred stock, k_p= 12%

Cost of equity, k_e = 17%

Proportion of debt, $w_d = \frac{6000,000}{16000,000} = 0.375$

Proportion of Preferred stock, $w_p = \frac{2000,000}{16000,000} = 0.125$

Proportion of equity, $w_e = \frac{8000.000}{16000,000} = 0.5$

Marginal tax rate, T = 40% = .40

Weighted average cost of capital (WACC),

$$k_a = w_d k_d (1 - T) + w_p k_p + w_e k_e$$
$$= 0.375\,(13\%)\,(1\text{-}.40) + 0.125\,(12\%) + 0.5\,(17\%)$$
$$= 2.925\% + 1.5\% + 8.5\%$$
$$= 12.93\%$$

9. Star Products Company is a growing manufacturer of automobile accessories whose stock is actively traded on the over-the-counter exchange. During 2003, the Dallas-based company experienced sharp increases in both sales and earnings. Because of this recent growth, Melissa Jen, the company's treasurer, wants to make sure that available funds are being used to their fullest. Management policy is to maintain the current capital structure proportions of 30% long-term debt, 10% preferred stock, and 60% common stock equity for at least the next 3 years. The firm is in the 40% tax bracket.

To estimate the firm's weighted average cost of capital (WACC), Ms. Jen contacted a leading investment banking firm, which provided the financing cost data shown in the following table.

Financing Cost Data

Star Products Company

Long-term debt: The firm can raise $450,000 of additional debt by selling 15-year, $1,000-par-value, 9% coupon interest rate bonds that pay *annual interest.* It expects to net $960 per bond after flotation costs. Any debt in excess of $450,000 will have a before-tax cost, k_d, of 13%.

Preferred stock: Preferred stock, regardless of the amount sold, can be issued with a $70 par value and a 14% annual dividend rate and will net $65 per share after flotation costs.

Common stock equity: The firm expects dividends and earnings per share to be $0.96 and $3.20, respectively, in 2004 and to continue to grow at a constant rate of 11% per year. The firm's stock currently sells for $12 per share. Star expects to have $1,500,000 of retained earnings available in the coming year. Once the retained earnings have been exhausted, the firm can raise additional funds by selling new common stock, netting $9 per share after underpricing and flotation costs.

Required:

a. Calculate the cost of each source of financing, as specified:

(1) Long-term debt, first $450,000.

(2) Long-term debt, greater than $450,000.

(3) Preferred stock, all amounts.

(4) Common stock equity, first $1,500,000.

(5) Common stock equity, greater than $1,500,000.

b. Find the break points associated with each source of capital, and use them to specify each of the ranges of total new financing over which the firm's weighted average cost of capital (WACC) remains constant.

c. Calculate the weighted average cost of capital (WACC) over each of the ranges of total new financing specified in part **b.**

Answer:

a. Cost of financing sources

(1) Debt: \$0 - \$450,000:

Given that,

$I = \$1{,}000 \times 9\% = \$90$

$N_d = \$960$

Now, $k_d = \dfrac{I + \frac{(\$1{,}000 - N_d)}{n}}{\frac{N_d + \$1{,}000}{2}}$

$= \dfrac{\$90 + \frac{(\$1{,}000 - \$960)}{n}}{\frac{\$960 + \$1{,}000}{2}}$

$= \dfrac{\$92.67}{\$980}$

$= 0.0946$

$= 9.46\%$

$k_{dt} = k_d\ (1 - t)$

$= 9.46\%\ (1 - .4)$

$= 5.7\%$

(2) Debt: Above \$450,000

$k_{dt} = k_d\ (1 - t)$

$= 13\%\ (1 - .4)$

$= 7.8\%$

(3) Preferred Stock:

Given that,

$D_p = \$70 \times 14\%$

$= \$9.80$

Now, $k_p = \dfrac{D_p}{P_0 - F}$

$= \dfrac{\$9.80}{\$65}$

$= 0.151$

$= 15.1\%$

(4) Common Stock Equity: \$0 - \$1,500,000

$$k_e = \frac{D_1}{P_0} + g$$
$$= \frac{\$0.96}{\$12} + .11$$
$$= .19 \text{ or } 19\%$$

(5) Common Stock Equity: Above $1,500,000

$$k_e = \frac{D_1}{P_0 - F} + g$$
$$= \frac{\$0.96}{\$9} + .11$$
$$= (\$.96 \div \$9) + .11$$
$$= .217 \text{ or } 21.7\%$$

b. Break Point = $\frac{\text{Available of source of capital}}{\text{Proportion of source of capital}}$

$$BP_{\text{Long-term debt}} = \frac{\$450,000}{.30} = \$1,500,000$$
$$BP_{\text{Common equity}} = \frac{\$1,500,000}{.60} = \$2,500,000$$

c. Weighted average cost of capital (WACC):

Type of Capital	Proportion of Capital	Cost of Capital	Weighted Cost
(1) From $0 to $1,500,000:			
Long-term debt	.30	5.7%	1.71%
Preferred stock	.10	15.1%	1.51%
Common stock equity	.60	19.0%	11.40%
		WACC	= 14.62%
(2) From $1,500,000 to $2,500,000:			
Long-term debt	.30	7.8%	2.34%
Preferred stock	.10	15.1%	1.51%
Common stock equity	.60	19.0%	11.40%
		WACC	= 15.25%
(3) Above $2,500,000:			
Long-term debt	.30	7.8%	2.34%
Preferred stock	.10	15.1%	1.51%
Common stock equity	.60	21.7%	13.02%
		WACC	= 16.87%

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Practice Questions

1. Carolynco Textile Systems has recently issued £20 million of 20-year bonds at £100 par value which are currently trading at £96 with a current yield of 14.67 per cent. The company currently has 4.5 million shares in issue, no new issues have been made and the current market price is quoted at £6.30. The cost of equity has been calculated at 15.5 per cent and corporation tax is 30 per cent. Determine the current weighted average cost of capital (WACC) for the company.

2. Details of the capital structure of Celtic Enterprises, a specialist printing company, are as follows:

	£m
Ordinary share capital (£1 ordinary shares fully paid)	5
6% Preference shares (£1 nominal value)	2
Retained earnings	14
7% Debentures (irredeemable) £100 par	1
8% Debentures (redeemable) £100 par	2

Ordinary shares are currently trading at £2.80 and a dividend of 34 pence per share is due for payment. Dividend per share has been growing at 3 per cent per annum and the directors expect this rate of growth to continue. Preference shares have a current market value of £0.89. The irredeemable debentures are trading at £92. The redeemable debentures are due for redemption in 4 years time and currently have a market value of £88. The rate of corporation tax is 33 per cent.

You are required to:

(a) Calculate the cost of each individual source of capital in the company's capital structure.

(b) Calculate the weighted average cost of capital (WACC).